
Supplementary Materials

Contracting Services for Response to Extreme Weather Events in Public Organizations

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This supplement includes:

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S2. Model Specification;

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S4. Common Method Variance Analysis;

S5. Correlation Matrix;

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Impacts Level;

S7. Dedicated Financial Resources across Extreme Weather Impacts Level;

S8. Perceived Severity as an Alternative Measurement for Extreme Weather
Event Impacts;

S9. Alternative Agency Aggregation;

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S1. Survey Sampling and Administration

In this section, we describe the criteria used to develop sample frame of the survey data and how the survey was administered.

The largest public transit agencies are the target population of this survey, which is defined as all public transit agencies operating fixed-route bus or/and rail transit systems in metropolitan areas with an annual fare revenue of at least one million dollars in 2017. The research team defined the sample frame by applying the following criteria.

(1) A list of public transit agencies serving urbanized area was drawn from National Transit Database (NTD). Residents living in large urban areas account for 96% of the passenger trips nationally (APTA, 2017). Their experience with extreme weather events matters most for the security and travel of most passengers in the U.S.

(2) The agencies whose annual fare revenue of over one million dollars in 2017 were kept. These largest transit agencies serving urbanized areas must report their operating data to the Federal Transit Administration. The FTA reporting requirements are waived for urban agencies that operate less than 30 vehicles as well as those that operate no services (e.g., transit planning agencies). Sampling the largest public transit agencies helps us to access their operating data from NTD for further analysis. Additionally, larger organizations have stronger capacity and more information, and, thus, are more likely to respond a survey than smaller agencies (Tomaskovic-Devey et al., 1994; Xiang and Chen, 2023). Targeting this group has the potential for increasing survey response rate and data quality.

(3) University-run agencies without fixed-route transit systems were removed.

Largest transit agencies with fixed-route transit system spend more effort on building infrastructure, maintaining complex transit system, and coping with complex management and operation problems. Complex transit systems make the agencies more vulnerable to the adverse impact of extreme weather events. The elements within their transit systems are tightly coupled (Perrow, 1994). Even small weather-caused breakdowns on one element may have a domino effect on their whole transit system. These transit system characteristics require agency to have a larger management team, enhanced capacity, and sufficient resources to respond to the changing external environment (i.e., extreme weather events) (Zhang et al., 2018).

(4) Private companies were removed as this survey focuses on public agencies.

For each of the identified agencies filtered from the previous criteria, the research team surveyed department heads (i.e., top managers in this study) in five common departments in each agency: operations, maintenance, engineering, service planning, and strategic planning.

There are several rationales of selecting top managers from these departments. First, they have a wealth of experience and expertise in operating and planning transit infrastructure, and they are familiarized with the local situation relevant with complex transit system in management and corresponding responses to extreme weather events. Second, facing the risk of extreme weather events, top managers are the critical decision-makers responsible for devising strategies to address and prevent adverse

impacts from changing, external environment in the time of emergency (Tabesh and Vera, 2020). Third, the experience and perspectives of individuals within organizations vary and depend on their expertise and positions within the organization (Schwenk, 1990; Zhang et al., 2018). Top managers' insights and experiences provide a comprehensive representation of the prevailing conditions, activities, and future directions of public transit agencies (Huang and Villadsen, 2023). Last, surveying multiple managers from one agency also improves possibility of having one respondent to report the actions and experience of the agency and reduces bias in the representativeness of responding agency.

The research team identified top managers and collected their contact information through several ways: (1) searching agency web sites; (2) making phone calls with staffs at public transit agencies; (3) submitting Freedom of Information Act requests to the agencies. Not all the agencies have all five departments, and some agencies were not willing to disclose contact information and participate in the study. The final sample frame includes 1,039 respondents from 291 agencies (Zhang and Welch, 2023).

Survey administration experienced pre-testing and full administration. The finalized survey instrument was coded as an online survey in Sawtooth Software®. The research team pretested the survey instruments with seven purposively selected top managers who were known to the team and who were willing to provide detailed comments and feedback on specific questions to maximize clarity. The research team revised the survey instruments with pretest respondents' constructive feedback. After

the pre-testing, the revised and finalized survey was administered online to the full sample frame. The full survey opened on April 9th, 2019 and ran until June 19th, 2019. The research team sent a hard-copy invitation letter to each of the respondents informing them of the survey, its aims and the reasons why they were invited. One week after sending the hard-copy letter, respondents were invited by email to participate in the study. Each email included an online survey link and a unique personal ID and password. Following the initial invitation, biweekly follow-up reminders were sent to each of the respondents who had not responded or had started but not completed the survey. Five reminders were sent from April 17th, 2019 to May 29th, 2019. At the end of the survey instrument, the research team provided an option for respondents to enter a lottery for a \$50 Amazon gift card.

Of the initial 1,039 individuals in the sample, several were removed during survey administration because they were either not reachable (N=73), unwilling to participate (N=26), not qualified to respond, or no longer employed by the agency (N=87). During the survey, several individuals were identified to supplement those replaced and invited to participate in the study. The total final valid sample size was 853 from 278 agencies.

Finally, 313 managers representing 194 transit agencies responded to the survey. The manager-level response rate is 38% (APPOR Response Rate 4) and the agency-level response rate is 70%. The manager-level response rate 4 was calculated following criteria developed by American Association for Public Opinion Research (2023).

S2. Model Specification

$$\ln \frac{E\{Contracting_i\}}{1 - E\{Contracting_i\}} = \beta_0 + \beta_1 Uncertainty_i + \beta_2 ContracingRatio_i + \beta_3 Finance_i + \gamma X_i + \varepsilon_i$$

$$\ln \frac{E\{Contracting_i\}}{1 - E\{Contracting_i\}} = \beta_0 + \beta_1 Uncertainty_i + \beta_2 Uncertainty_i^2 + \beta_3 ContracingRatio_i + \beta_4 Finance_i + \gamma X_i + \varepsilon_i$$

In both equations, i refers to the agency. $Contracting_i$ represents whether the agency has contracted other organizations for immediate emergency responses or long-term emergency planning. $Uncertainty_i$ refers to the linear term of agency i 's perception of extreme weather event impact, while $Uncertainty_i^2$ refers to the squared term. $ContracingRatio_i$ refers to the reliance of agency i on using contracting to provide daily transit services. $Finance_i$ refers to whether the agency i has received dedicated financial resources for immediate emergency responses or long-term emergency planning. We include additional explanatory variables X_i as controls in the estimation. ε_i is the random error term.

Table S2.1. Distribution of manager-level observation

Number of managers representing the same agency	Number of agency- level observation	Total manager-level observations
6	1	6
5	0	0
4	4	16
3	19	57
2	64	128
1	106	106
Total	194	313

Table S2.2. Number of dropped agencies for modeling and associated missing values

Number of dropped agencies	Variables with missing values
5	Contracting for immediate emergency response services, Contracting for long-term emergency planning services
1	Contracting for immediate emergency response services, Contracting for long-term emergency planning services, Organizational uncertainty about extreme weather events
1	Reliance on contracting, Service area size, Operating expense of purchased services
1	Organizational uncertainty about extreme weather events, Dedicated financial resources, Extreme weather impacts

S3. Within-group Interrater Reliability (RWG)

Table S3.1 RWG Statistics.

Variables	Number of Agencies	Mean SD	Median RWG	% of Agencies, RWG > 0.5	% of Agencies, RWG > 0.7	% of Agencies, RWG > 0.9
Contracting for Immediate Emergency Response	82	0.31	1	53.7%	53.7%	53.7%
Contracting for Long-term Planning	82	0.16	1	75.6%	75.6%	75.6%
Uncertainty about Extreme Weather Events	87	0.67	0.75	79.3%	65.5%	26.4%
Dedicated financial resources for immediate emergency responses	86	0.12	1	81.4%	81.4%	81.4%
Dedicated financial resources for long-term planning	86	0.06	1	90.7%	90.7%	90.7%
Extreme weather impacts	86	1.32	0.83	88.4%	77.9%	44.2%

Table S3.2 Variance Statistics.

Variables	Number of Agencies	Mean Within-Agency SD	Between-Agency SD
Uncertainty about Extreme Weather Events (perceptual variable)	87	0.68	0.61
Extreme weather impacts (continuous variable)	86	1.32	1.70

S4. Common Method Variance Analysis

We conducted common method variance analysis on manager-level survey responses. Table S4.1 shows the result of Harman's single-factor test. No single factor explains more than 20 percent of the covariance among all survey items. This indicates that common method variance (CMV) is not a significant concern for our analysis. Table S4.2 shows the test result of unmeasured latent method construct (ULMC) method. We construct models with ULMC and without ULMC by using confirmatory factor analysis (CFA) then compared the factor loading of all observed survey items. If the factor loading differences between models with ULMC and without ULMC is below 0.2, CMV is not a concern in the regression model with these measurement items (Afthanorhan et al., 2021; Fusi et al., 2023). The small differences of factor loading in our models are all below 0.2, which indicates that CMV is not an issue for our regression results.

Table S4.1. Harman's single-Factor Test Results.

Principal Components	Eigenvalues		
	Proportion of Variance (%)	Cumulative Proportion of Variance (%)	Standard Deviation
1	16.49	16.49	1.82
2	13.78	30.27	1.66
3	10.30	40.57	1.44
4	9.10	49.67	1.35
5	5.99	55.66	1.09
6	5.83	61.49	1.08
7	5.11	66.60	1.01

Note: The Principal Component Analysis (PCA) method was utilized to extract principal components in the Harman's single-Factor Test. A total of 21 principal components were extracted initially. We only reported the components whose proportions of variance are larger than 5%. None of the first seven principal components individually account for more than 20% of the overall variability of our measurement items. This result shows that the CMV is does not pose a significant concern for our analysis.

1 Table S4.2. Factor Loading Comparison for Measurement Items.

Latent Variable	Measurement Items	Factor Loading without ULMC	Factor Loading with ULMC	Difference
Contracting for emergency responses Contracting for emergency Planning	<i>Over the past two years, has your agency done or is it currently doing any of the following to address extreme weather? (1=Yes, 0=No or I don't know)</i>			
	Contracted other organizations to provide assistance during extreme weather	1	1	0
	Contracted experts to help your agency plan for extreme weather	1	1	0
Uncertainty about extreme weather events	<i>To what extent do the following factors limit your agency's ability to build capacity to respond effectively to extreme weather? (1=Not at all, 2=Low extent, 3=Moderate extent, 4=High extent, 5=Very high extent)</i>			
	(1) Uncertainty about the likelihood of extreme weather events			
	(2) Uncertainty about the impacts of extreme weather events	.694	.636	.058
	(3) Uncertainty about best options available to address extreme weather events			
Dedicated financial resources	<i>During the last two years, did the following happen to your agency as a result of extreme weather events? (1=Yes, 0=No or I don't know)</i>			
	My agency received funding to plan for future extreme weather	.150	.262	.112
	My agency received funding to repair or replace assets damaged by extreme weather			
Extreme weather impacts	<i>During the last two years, did the following happen to your agency as a result of extreme weather events? ". A list of survey items includes the potential negative result of extreme weather events (1=Yes, 0=No or I don't know)</i>			
	(1) Excessive delays in transit services			
	(2) Temporary shutdown of all transit services			
	(3) Temporary partial shutdown of some transit services			
	(4) Failure of portions of transit services or systems			
	(5) Damage to infrastructure or facilities			
	(6) Damage to vehicles or equipment			
	(7) Passenger or operator injuries			
	(8) Passenger or operator loss of life	.540	.589	.049
	(9) Other injuries or loss of life (e.g., bystanders)			
	(10) Lawsuits were filed against my agency			
	(11) Political oversight of my agency has increased			
	(12) Individuals in my agency were removed, terminated, forced to resign or voluntarily resigned			
	(13) My agency was the target of negative publicity			

2 Note: We employed Confirmatory Factory Analysis (CFA) to construct models with ULMC and without ULMC for the latent variables that were included in our regression models. To maintain consistency with our
3 measurement approach for the key constructs in the regression model, we constrained all the factor loadings of measurement items for their respective latent variables to be equal. The difference of factor loading estimate
4 between models with ULMC and without ULMC are small. All the differences are below the threshold of 0.2, which is used to infer the presence of a common method variance influencing the association between
5 constructs in the statistical models. Thus, evidence from table S4,2 indicates that the common method variance is not a significant concern for the result of our regression model.

S5. Correlation Matrix

Table S5. Correlation Matrix.

Key variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Contracting for immediate emergency responses	1															
2. Contracting for long-term emergency planning	0.379***	1														
3. Uncertainty about extreme weather events	0.032	-0.024	1													
4. Reliance on contracting	0.055	0.187*	0.009	1												
5. Dedicated financial resources for immediate emergency responses	0.325***	0.11	0.071	-0.009	1											
6. Dedicated financial resources for long-term planning	0.263***	0.353**	0.172*	0.072	0.405**	1										
7. Extreme weather impacts	0.234***	0.079	0.118	-0.129	0.34**	0.175*	1									
8. Organizational size (log)	0.07	-0.014	0.029	-0.258**	0.144*	-0.069	0.116	1								
9. Service area size (log sq miles)	0.131	0.126	-0.183*	0.094	0.193*	0.151*	0.244**	0.044	1							
10. Service efficiency	0.16*	0.041	0.096	0.041	0.081	0.122	0.121	-0.066	0.103	1						
11. Service effectiveness	0.112	-0.1	0.057	-0.12	0.119	0.012	0.002	0.122	-0.204**	0.316**	1					
12. Local funding percentage	0.095	-0.039	-0.005	0.033	0.077	0.033	0.017	0.16*	-0.047	-0.245**	0.052	1				
13. State funding percentage	-0.047	-0.009	0.085	0.063	-0.041	0.023	0.022	-0.2**	-0.134	0.119	0.095	-0.273**	1			
14. Federal funding percentage	-0.022	-0.035	0.066	0.054	-0.007	-0.114	0.003	-0.183*	-0.173*	-0.137	-0.004	-0.017	-0.012	1		
15. Operating expense of purchased services (per capital \$)	0.061	0.156*	0.085	0.732**	-0.009	0.015	-0.062	-0.147*	0.084	0.008	0.007	-0.015	0.103	-0.012	1	
16. Political environment	0.171*	0.121	0.114	0.206**	0.116	0.158*	0.043	0.186*	0.128	0.164*	0.123	0.165*	-0.012	-0.372**	0.258**	1
17. Department: Strategic Planning	-0.021	0.002	-0.064	0.194*	-0.057	-0.021	-0.155*	0.031	-0.19**	0.061	0.056	-0.019	0.088	-0.037	0.096	0.019

Note: Contracting for assistance, contracting for planning, dedicated financial resources for immediate emergency responses, and dedicated financial resources for long-term planning are binary variables (1=Yes, 0=No). Responses on the latent variables of uncertainty about extreme weather event were provided on a five-point Likert scale (1=Not at all, 2=Low extent, 3=Moderate extent, 4=High extent, 5=Very high extent). * $p < .05$, ** $p < .01$, *** $p < .005$.

S6. Uncertainty about Extreme Weather Events across Extreme Weather Event Impacts Level

Figure S6 shows the uncertainty level of three survey items across the three groups. According to the mean and frequency distribution of the level of extreme weather impacts, we classified the respondents into three groups. As the mean is 3.27, we categorize the respondents who experienced more than 4 types of impacts as the “High” group, while respondents who experienced 0 to 4 types of extreme weather impacts are categorized as the “Medium” group. Respondents who reported no impact of extreme weather are categorized as the “Low” group. The percentages for these three groups are correspondingly 27.9%, 64.8%, and 7.3%. Our ANOVA test indicates a statistically significant difference in agency uncertainty about the impacts of extreme weather events and best options to address them, based on the level of extreme weather impacts experienced ($p < 0.05$).

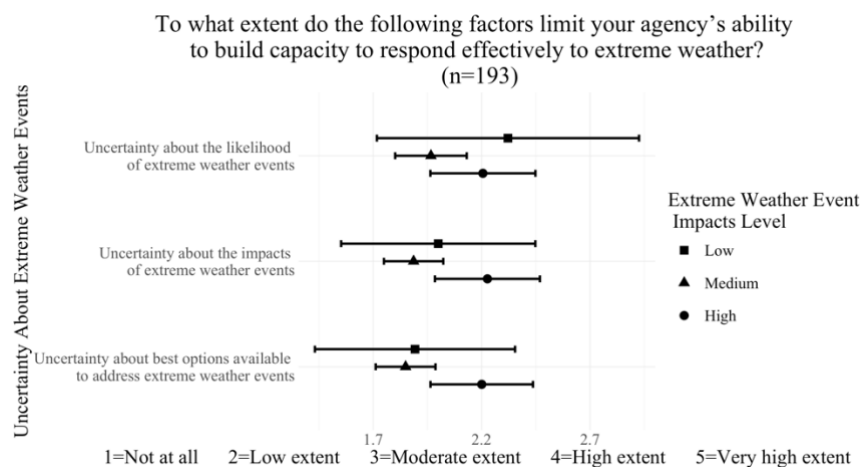


Figure S6. Uncertainty about Extreme Weather Events across Extreme Weather Event Impacts Level.

Note: As the mean of extreme weather impacts is 3.27, we categorize the agencies whose level is higher than 4 as the “High” group, the agencies whose level is from 0 to 4 as the “Medium” group, and agencies whose level is 0 as the “Low” group.

S7. Dedicated Financial Resources across Extreme Weather Event Impacts Level

Figure S7 shows the percentage of agencies that received dedicated financial resources for extreme weather across three groups with different extreme weather impacts level. The ANOVA test indicates a statistically significant difference in public transit agency receipt of funding to repair damaged assets ($p < 0.005$) and to plan for the future, based on the level of extreme weather impacts experienced ($p < 0.05$).

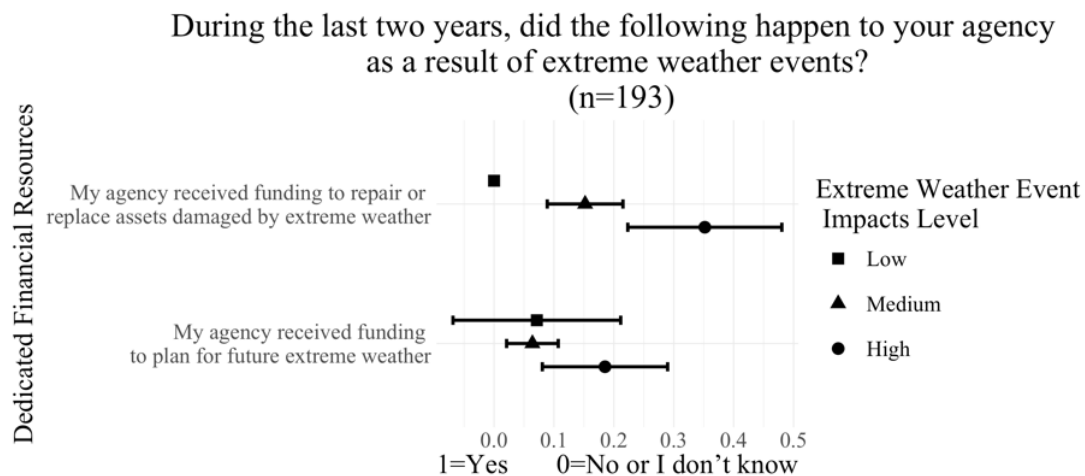


Figure S7. Dedicated Financial Resources across Extreme Weather Event Impacts Level.

Note: As the mean of extreme weather impacts is 3.27, we categorize the agencies whose level is higher than 4 as the “High” group, the agencies whose level is from 0 to 4 as the “Medium” group, and agencies whose level is 0 as the “Low” group.

S8. Perceived Severity as an Alternative Measurement for Extreme Weather Event Impacts

We conduct a robustness check by using the perceived severity of extreme weather events as an alternative for the measurement of extreme weather impacts. Following Zhang et al. (2018), we measure perceived severity of extreme weather events using a scale calculated from the mean of responses for the following survey items. “Considering the extreme weather events that have happened in your area in the previous two years, has the level of adverse impact been catastrophic, major, moderate, minor, or none?” (1=minor, 5=catastrophic). Survey items include 12 types of extreme weather events: (1) Extreme cold temperatures, (2) Extreme heat wave, (3) Wildfires, (4) River floods, (5) Flash floods, (6) Hurricanes or tropical storms, (7) Severe rainstorms or thunderstorms, (8) Storm surges, (9) Extreme high winds, (10) Tornadoes touchdown, (11) Extreme snow storms, (12) Freezing rain/ice.

Table S8 presents the robustness check results, comparing models with alternative measurements to the original models. Models 1a and 2a include the original measurement of extreme weather impacts. Models 1b and 2b include the alternative measurement of perceived severity of extreme weather events. The significance and association of uncertainty, reliance on contracting, and recipient of dedicated financial resources on contracting for both immediate agency emergency response and long-term emergency planning remain consistent in the models with alternative measurement of extreme weather impacts. The goodness of fit indexes (i.e., Pseudo R-squared) are similar between original models and alternative models.

Interestingly, the original measurement of extreme weather impacts has a significant association with contracting for immediate emergency responses, while the alternative measurement does not (see model 1a and model 1b). This implies that public organizations are more likely to respond to the adverse outcomes brought by tangible impacts from extreme weather than by general perceived extreme weather severity.

Table S8. Robustness Check Results for Extreme Weather Severity

Explanatory variables	Contracting for immediate emergency responses		Contracting for long-term emergency planning	
	Quadratic Model 1a	Quadratic Model 1b	Quadratic Model 2a	Quadratic Model 2b
Uncertainty	0.794** (0.310)	0.784** (0.311)	0.114 (0.217)	0.09 (0.212)
Uncertainty squared	-0.195*** (0.073)	-0.19*** (0.073)	-0.041 (0.051)	-0.034 (0.049)
Reliance on contracting	-0.042 (0.150)	-0.082 (0.148)	0.190* (0.109)	0.179* (0.109)
Dedicated financial resources for immediate emergency responses	0.352*** (0.106)	0.389*** (0.1)		
Dedicated financial resources for long-term emergency planning			0.493*** (0.140)	0.535*** (0.134)
Control variables				
Extreme weather impacts	0.037* (0.022)		0.008 (0.016)	
Perceived extreme weather severity		0.04 (0.073)		-0.065 (0.055)
Organizational size	-0.005 (0.027)	-0.004 (0.027)	0.019 (0.020)	0.025 (0.02)
Service area size	0.015 (0.035)	0.022 (0.036)	-0.002 (0.026)	0.005 (0.025)
Service efficiency	0.533 (0.393)	0.65* (0.395)	-0.096 (0.269)	-0.094 (0.267)
Service effectiveness	0.003 (0.481)	-0.024 (0.483)	-0.565 (0.467)	-0.564 (0.468)
Local funding percentage	0.243 (0.202)	0.285 (0.204)	-0.202 (0.143)	-0.205 (0.145)
State funding percentage	0.045 (0.225)	0.089 (0.223)	-0.080 (0.160)	-0.061 (0.158)
Federal funding percentage	0.246 (0.391)	0.245 (0.396)	0.016 (0.288)	0.081 (0.29)
Operating expense of purchased services	0.001 (0.003)	0.002 (0.003)	-0.001 (0.002)	-0.001 (0.002)
Political environment	0.639* (0.355)	0.585* (0.355)	0.316 (0.263)	0.318 (0.266)
Department: Strategic Planning	0.003 (0.084)	-0.010 (0.084)	-0.020 (0.064)	-0.032 (0.063)
McFadden's Pseudo R-squared	0.155	0.142	0.167	0.172
Log Likelihood	-104.915	-106.091	-81.674	-81.000
AIC	241.830	244.182	195.348	194.001
Observations	186	185	186	185

Note: Coefficients are AME. Standard errors in parentheses; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

S9. Alternative Agency Aggregation

Table S9.1. Descriptive Statistics for Alternative Aggregation.

Variables	N	Mean	SD	Min	Max
Contracting for immediate emergency responses	127	0.20	0.41	0	1
Contracting for long-term emergency planning	127	0.14	0.35	0	1
Uncertainty about extreme weather events	131	2.06	0.87	1	4.33
Reliance on contracting	132	0.39	0.40	0	1
Dedicated financial resources for immediate emergency responses	132	0.14	0.34	0	1
Dedicated financial resources for long-term planning	132	0.08	0.27	0	1
Extreme weather impacts	132	3.25	2.23	0	11
Organizational size (log)	127	3.73	1.67	0	8.52
Service area size (log sq miles)	132	5.39	1.29	2.71	8.69
Service efficiency	133	0.20	0.13	0	0.98
Service effectiveness	133	0.16	0.09	0	0.64
Local funding percentage	133	0.27	0.24	0	0.79
State funding percentage	133	0.22	0.21	0	0.81
Federal funding percentage	133	0.17	0.12	0	0.52
Operating expense of purchased services (per capital \$)	132	14.67	20.42	-7.33	96.45
Political environment	133	0.55	0.13	0.30	0.88
Department: Strategic Planning	133	0.32	0.47	0	1

Note: Contracting for immediate emergency response services, contracting for long-term emergency planning services, dedicated financial resources for immediate emergency responses, and dedicated financial resources for long-term emergency planning are binary variables (1=Yes, 0=No or I don't know). Responses on the latent variables of uncertainty about extreme weather event were provided on a five-point Likert scale (1=Not at all, 2=Low extent, 3=Moderate extent, 4=High extent, 5=Very high extent). We include agencies with either a single respondent or with multiple respondents who provided consistent responses on the binary variables. Manager responses for other survey measures are averaged to the agency level.

Table S9.2. RWG Statistics for Reduced Sample.

Variables	Number of Agencies	Mean SD	Median RWG	% of Agencies, RWG > 0.5	% of Agencies, RWG > 0.7	% of Agencies, RWG > 0.9
Uncertainty about Extreme Weather Events	27	0.617	0.89	85.2%	70.4%	33.3%
Extreme weather impacts	27	1.10	0.89	96.3%	88.9%	48.1%

Table S9.3. Robustness Check Results: Probability That a Public Transit Agency Contract.

Explanatory variables	Contracting for immediate emergency responses		Contracting for long-term emergency planning	
	Full Quadratic Model 1	Reduced Quadratic Model 1	Full Quadratic Model 2	Reduced Quadratic Model 2
Uncertainty	0.794** (0.310)	0.478** (0.221)	0.114 (0.217)	0.187 (0.128)
Uncertainty squared	-0.195*** (0.073)	-0.095* (0.049)	-0.041 (0.051)	-0.046 (0.03)
Reliance on contracting	-0.042 (0.150)	0.024 (0.144)	0.190* (0.109)	0.163** (0.082)
Dedicated financial resources for immediate emergency responses	0.352*** (0.106)	0.22 (0.176)		
Dedicated financial resources for long-term emergency planning			0.493*** (0.140)	0.441 (0.327)
Control variables				
Extreme weather impacts	0.037* (0.022)	0.011 (0.017)	0.008 (0.016)	0.004 (0.009)
Organizational size	-0.005 (0.027)	0.012 (0.021)	0.019 (0.020)	0.026* (0.014)
Service area size	0.015 (0.035)	0.049 (0.033)	-0.002 (0.026)	0.019 (0.015)
Service efficiency	0.533 (0.393)	-0.254 (0.414)	-0.096 (0.269)	0.044 (0.163)
Service effectiveness	0.003 (0.481)	0.251 (0.498)	-0.565 (0.467)	-0.376 (0.29)
Local funding percentage	0.243 (0.202)	0.164 (0.18)	-0.202 (0.143)	-0.042 (0.086)
State funding percentage	0.045 (0.225)	0.076 (0.201)	-0.080 (0.160)	0.132 (0.094)
Federal funding percentage	0.246 (0.391)	0.063 (0.343)	0.016 (0.288)	0.05 (0.151)
Operating expense of purchased services	0.001 (0.003)	0.001 (0.003)	-0.001 (0.002)	-0.0 (0.001)
Political environment	0.639* (0.355)	0.423 (0.309)	0.316 (0.263)	-0.094 (0.141)
Department: Strategic Planning	0.003 (0.084)	-0.105 (0.071)	-0.020 (0.064)	-0.08** (0.041)
McFadden's Pseudo R-squared	0.155	0.210	0.167	0.415
Log Likelihood	-104.915	-50.492	-81.674	-30.156
AIC	241.830	132.984	195.348	92.311
Observations	186	125	186	125

Note: Coefficients are AME. Standard errors in parentheses; * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. In reduced models, we include agencies with either a single respondent or with multiple respondents who provided consistent responses on the binary variables. Manager responses for other survey measures are averaged to the agency level.

Contracting extreme weather event responses

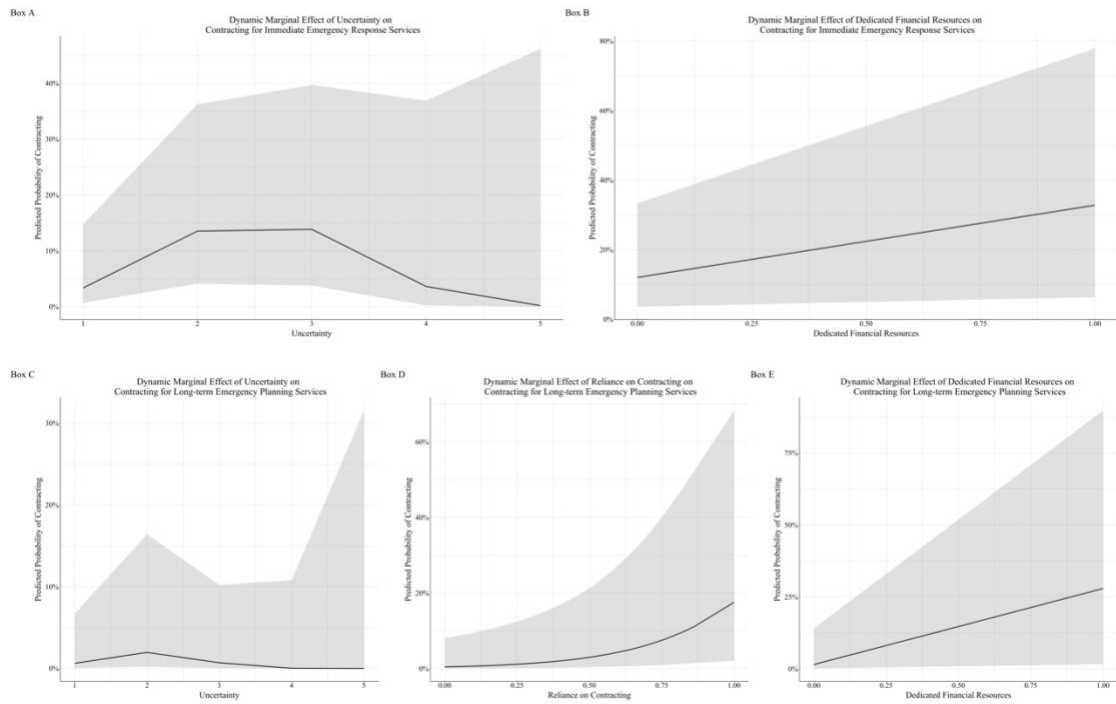


Figure S9. Robustness Check for Dynamic Marginal Effects of Significant Explanatory Variables.

Note: All other variables in the models are controlled at means. Shaded bands denote 95% confidence intervals.

S10. Moderating Effect of Extreme Weather Impact

Table S10.1. Logistic Model Results: Moderating Effect of Extreme Weather Impacts on Contracting for Immediate Emergency Responses.

Explanatory Variables	Contracting for Immediate Emergency Responses		
	Quadratic Model 1	Quadratic Model 2	Quadratic Model 3
Uncertainty	0.861*** (0.316)	0.796** (0.311)	0.821*** (0.317)
Uncertainty squared	-0.183** (0.072)	-0.195*** (0.073)	-0.203*** (0.075)
Reliance on contracting	-0.042 (0.151)	-0.05 (0.223)	-0.044 (0.151)
Dedicated financial resources on immediate emergency responses	0.354*** (0.107)	0.352*** (0.106)	0.146 (0.27)
Uncertainty: Extreme weather impacts	-0.03 (0.026)		
Reliance on contracting: Extreme weather impacts		0.002 (0.053)	
Dedicated financial resources on immediate emergency responses: Extreme weather impacts			0.048 (0.056)
Control variables			
Extreme weather impacts	0.106* (0.064)	0.036 (0.03)	0.028 (0.024)
Organizational size	-0.007 (0.027)	-0.005 (0.027)	-0.005 (0.027)
Service area size	0.016 (0.036)	0.015 (0.035)	0.012 (0.036)
Service efficiency	0.501 (0.396)	0.534 (0.394)	0.586 (0.401)
Service effectiveness	0.016 (0.483)	0.005 (0.483)	-0.035 (0.484)
Local funding percentage	0.232 (0.203)	0.243 (0.202)	0.231 (0.202)
State funding percentage	0.073 (0.228)	0.045 (0.225)	0.007 (0.23)
Federal funding percentage	0.219 (0.394)	0.246 (0.392)	0.26 (0.391)
Operating expense of purchased services	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)
Political environment	0.647* (0.358)	0.638* (0.356)	0.649* (0.357)
Department: Strategic Planning	0.01 (0.085)	0.004 (0.084)	0.004 (0.085)
McFadden's Pseudo R-squared	0.16	0.155	0.158
Log Likelihood	-104.236	-104.914	-104.525
AIC	242.471	243.828	243.051
Observations	186	186	186

Note: Coefficients are AME. Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$; *** $p < 0.01$.

Table S10.2. Logistic Model Results: Moderating Effect of Extreme Weather Impacts on Contracting for Long-term Emergency Planning.

Explanatory Variables	Contracting for Long-term Emergency Planning		
	Quadratic Model 1	Quadratic Model 2	Quadratic Model 3
Uncertainty	0.169 (0.228)	0.126 (0.22)	-0.037 (0.05)
Uncertainty squared	-0.04 (0.049)	-0.043 (0.051)	0.102 (0.216)
Reliance on contracting	0.187* (0.109)	0.11 (0.161)	0.195* (0.11)
Dedicated financial resources on long-term emergency planning	0.52*** (0.142)	0.491*** (0.141)	0.64*** (0.202)
Uncertainty: Extreme weather impacts	-0.017 (0.02)		
Reliance on contracting: Extreme weather impacts		0.026 (0.039)	
Dedicated financial resources on long-term emergency planning : Extreme weather impacts			-0.026 (0.034)
Control variables			
Extreme weather impacts	0.046 (0.047)	-0.004 (0.023)	0.015 (0.018)
Organizational size	0.018 (0.02)	0.019 (0.02)	0.021 (0.02)
Service area size	-0.003 (0.026)	-0.002 (0.026)	-0.003 (0.026)
Service efficiency	-0.12 (0.269)	-0.086 (0.272)	-0.12 (0.268)
Service effectiveness	-0.588 (0.471)	-0.57 (0.472)	-0.551 (0.469)
Local funding percentage	-0.207 (0.144)	-0.197 (0.144)	-0.207 (0.144)
State funding percentage	-0.069 (0.16)	-0.08 (0.16)	-0.066 (0.16)
Federal funding percentage	-0.005 (0.288)	-0.002 (0.292)	0.01 (0.289)
Operating expense of purchased services	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
Political environment	0.308 (0.262)	0.305 (0.264)	0.319 (0.263)
Department: Strategic Planning	-0.018 (0.064)	-0.018 (0.064)	-0.027 (0.064)
McFadden's Pseudo R-squared	0.171	0.170	0.170
Log Likelihood	-81.310	-81.446	-81.384
AIC	196.620	196.892	196.768
Observations	186	186	186

Note: Coefficients are AME. Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$; *** $p < 0.01$.

References

- Afthanorhan, Asyraf, Zainudin Awang, Norliana A. Majid, Hazimi Foziah, Izzat Ismail, Hussam A. Halbusi, and Shehnaz Tehseen. 2021. "Gain More Insight From Common Latent Factor in Structural Equation Modeling." *Journal of Physics: Conference Series* 1793 (012030): 1-9. <https://doi.org/10.1088/1742-6596/1793/1/012030>.
- American Association for Public Opinion Research. 2023. *Standard Definitions: Final Dispositions of Case Codes and Outcome Rates for Surveys*, 10th ed. American Association for Public Opinion Research. <https://aapor.org/wp-content/uploads/2023/05/Standards-Definitions-10th-edition.pdf>.
- APTA (American Public Transportation Association). 2017. "Who Rides Public Transportation." Accessed September 24, 2024. <https://www.apta.com/wpcontent/uploads/Resources/resources/reportsandpublications/Documents/APTA-Who-Rides-Public-Transportation-2017.pdf>.
- Fusi, Federica, Fengxiu Zhang, and Eric W. Welch. 2023. "Intra-Organizational Communication in Public Agencies: The Effects of Contracting Out Core Services." *The American Review of Public Administration* 53 (5-6): 224-242. <https://doi.org/10.1177/02750740231162346>.
- Huang, Ting, and Anders R. Villadsen. 2023. "Top Managers in Public Organizations: A Systematic Literature Review and Future Research Directions." *Public Administration Review* 83 (6): 1618-1634. <https://doi-org.ezproxy1.lib.asu.edu/10.1111/puar.13628>.
- Perrow, Charles. 1994. "The Limits of Safety: The Enhancements of a Theory of Accidents." *Journal of Contingencies & Crisis Management* 2 (4): 212-220. <https://doi-org.ezproxy1.lib.asu.edu/10.1111/j.1468-5973.1994.tb00046.x>.
- Schwenk, Charles R. 1990. "Conflict in Organizational Decision Making: An Exploratory Study of Its Effects in For-Profit and Not-for-Profit Organizations." *Management Science* 36 (4): 436-448. <https://www.jstor.org/stable/2632008>.
- Tabesh, Pooya, and Dusya M. Vera. 2020. "Top Managers' Improvisational Decision-Making in Crisis: A Paradox Perspective." *Management Decision* 58 (10): 2235-2256. <https://doi.org/10.1108/MD-08-2020-1060>.
- Tomaskovic-Devey, Donald, Jeffrey Leiter, and Shealy Thompson. 1994. "Organizational Survey Nonresponse." *Administrative Science Quarterly* 39 (3): 439-457. <https://www.jstor.org/stable/2393298>.
- Xiang, Tianyi, and Yifan Chen. 2023. "To Coordinate or Not? A Configurational Approach to Understand Public Organizations' Emergency Preparedness Coordination." *Public Administration Review* 84 (5): 817-832. <https://doi-org.ezproxy1.lib.asu.edu/10.1111/puar.13752>.
- Zhang, Fengxiu, Eric W. Welch, and Qing Miao. 2018. "Public Organization Adaptation to Extreme Events: Mediating Role of Risk Perception." *Journal of Public Administration Research and Theory* 28 (3): 371-387. <https://doi.org/10.1093/jopart/muy004>.

- 1 Zhang, Fengxiu, and Eric E. Welch. 2023. "Explaining Public Organization Adaptation
- 2 to Climate Change: Configurations of Macro-and Meso-Level Institutional
- 3 Logics." *Journal of Public Administration Research and Theory* 33 (2): 357-374.
- 4 <https://doi.org/10.1093/jopart/muac027>.